

ORF307: Optimization

Precept: #9

Agenda

- Optimality conditions and Complementarity
- Sensitivity analysis
 - Adding a new variable
 - Adding a new constraint Global/Local sensitivity analysis
 - Example
- Farkas lemma

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Optimality Conditions & Complementary Slackness

A **feasible primal solution** x^* and a **feasible dual solution** y^* are **optimal for their respective problems** if and only if their **Duality Gap is zero**.

Specifically, for the Standard Form LP

$$\min c^T x \quad s.t. \quad Ax = b, x \geq 0$$

and its Dual

$$\max -b^T y \quad s.t. \quad A^T y + c \geq 0$$

the following three conditions are necessary and sufficient for $\mathbf{x}^*$ and $\mathbf{y}^*$ to be optimal:

1. x^* is **Primal Feasible**: $Ax^* = b, \quad x^* \geq 0$
2. y^* is **Dual Feasible**: $A^T y^* + c \geq 0$

3. Complementary Slackness (CS) holds:

$$(A^T y^* + c)_j \cdot x_j^* = 0, \quad \text{for } j = 1, \dots, n$$

Since $A^T y^* + c$ (the dual slack vector) and x^* are both non-negative vectors, this is equivalent to:

$$(A^T y^* + c)^T x^* = 0$$

Intuition: If a variable is positive ($x_j^* > 0$), its corresponding dual constraint must be tight (slack is zero). If a dual constraint is non-tight (slack > 0), its corresponding primal variable must be zero.

Primal vs. Dual Simplex Behavior

The two main Simplex algorithms are designed to **maintain** certain optimality conditions while moving toward satisfying the others.

	Primal Feasibility	Dual Feasibility	Duality Gap
Primal Simplex	✓ (yes)	× (violates until optimal)	✓ (yes)
Dual Simplex	× (violates until optimal)	✓ (yes)	✓ (yes)

Proof that the Duality Gap is maintained at zero ($c^T x + b^T y = 0$):

In both Simplex methods, we work with a Basic Solution x and the corresponding implicit dual vector $y = -(A_B^{-1})^T c_B$ (or $y = -c_B^T A_B^{-1}$).

1. **Primal Cost** ($c^T x$): Since x is a basic solution, all non-basic variables x_N are zero. Thus:

$$c^T x = c_B^T x_B + c_N^T x_N = c_B^T x_B$$

2. **Dual Cost** ($-b^T y$): We substitute $y = -(A_B^{-1})^T c_B$ which can also be written as $y = -A_B^{-T} c_B$ (since the transpose of A^{-1} can also be denoted as A^{-T}):

$$-b^T y = -b^T (-A_B^{-T} c_B) = (A_B^{-1} b)^T c_B = x_B^T c_B$$

3. **Duality Gap:** Substituting the above two equations into the duality gap yields:

$$c^T x - (-b^T y) = c_B^T x_B - x_B^T c_B = 0$$

The duality gap is zero because the current solution x and the implicitly derived dual solution y are constructed from the same basis B .

Sensitivity analysis: adding a new variable

Consider the LP in standard form:

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

where $c, x \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$.

Add a new non-negative variable x_{n+1} :

$$\begin{aligned} \min_x \quad & c^T x + c_{n+1} x_{n+1} \\ \text{s.t.} \quad & Ax + A_{n+1} x_{n+1} = b \\ & x \geq 0, x_{n+1} \geq 0 \end{aligned} \tag{1}$$

where $A_{n+1} \in \mathbb{R}^m$ is an m -dimensional column vector.

If (x^*, y^*) is an optimal primal/dual solution pair, then $(x^*, 0)$ is also primal feasible.

We can write the LP problem (1) in compact form

$$\begin{aligned} \min_{\hat{x}} \quad & \hat{c}^T \hat{x} \\ \text{s.t.} \quad & \hat{A} \hat{x} = \hat{b} \\ & \hat{x} \geq 0 \end{aligned} \tag{2}$$

where

$$\hat{c} = \begin{pmatrix} c \\ c_{n+1} \end{pmatrix} \in \mathbb{R}^{n+1} \quad \hat{x} = \begin{pmatrix} x \\ x_{n+1} \end{pmatrix} \in \mathbb{R}^{n+1}, \quad \hat{A} = \begin{bmatrix} A & A_{n+1} \end{bmatrix} \in \mathbb{R}^{m \times (n+1)}, \quad \hat{b} = b$$

The **dual** problem of the previous LP is defined as

$$\begin{aligned} \max_{\hat{y}} \quad & -\hat{b}^T \hat{y} \\ \text{s.t.} \quad & \hat{A}^T \hat{y} + \hat{c} \geq 0 \end{aligned}$$

which can be expanded as follows

$$\begin{aligned} \max_y \quad & -b^T y \\ \text{s.t.} \quad & A^T y + c \geq 0 \\ & A_{n+1}^T y + c_{n+1} \geq 0 \end{aligned}$$

- If y^* satisfies the new dual constraint, i.e., $A_{n+1}^T y^* + c_{n+1} \geq 0$:

This means the reduced cost of the new variable, $\bar{c}_{n+1}$, is non-negative ($\bar{c}_{n+1} \geq 0$).

1. The solution $(x^*, 0)$ is primal feasible (by construction).
2. The dual solution y^* is dual feasible (since all constraints, including the new one, are satisfied).

3. Duality Gap is Zero: Since $x_{n+1} = 0$, the complementary slackness condition for the new variable is trivially satisfied ($\bar{c}_{n+1} \cdot x_{n+1} = \bar{c}_{n+1} \cdot 0 = 0$). All original variables and constraints maintain complementary slackness because (x^*, y^*) was originally optimal.

Since both primal and dual solutions are feasible and satisfy complementary slackness (meaning the duality gap is zero), the solution $(x^*, 0)$ is optimal for the new LP.

- Otherwise ($A_{n+1}^T y^* + c_{n+1} < 0$):

We have a solution that is primal feasible but dual infeasible. Therefore, we can run the **Primal Simplex**, starting from the initial Basic Feasible Solution $(x^*, 0)$, to bring the improving variable x_{n+1} into the basis and find the new optimal solution.

Sensitivity analysis: adding a new constraint

Let us now investigate what happens when we add an extra constraint to our original LP

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & a_{m+1}^T x = b_{m+1} \\ & x \geq 0 \end{aligned}$$

The dual problem is:

$$\begin{aligned} \max_{y, y_{m+1}} \quad & -b^T y - b_{m+1} y_{m+1} \\ \text{s.t.} \quad & A^T y + a_{m+1} y_{m+1} + c \geq 0 \end{aligned}$$

We can easily derive that $(y^*, 0)^T$ is dual feasible. Then we have the two cases:

- If $a_{m+1}^T x^* = b_{m+1}$: The original optimal solution x^* satisfies the new constraint, meaning it is **still primal feasible** in the expanded problem. Furthermore, the dual solution y^* of the original dual problem, augmented with $y_{m+1} = 0$ becomes $(y^*, 0)^T$ and **remains dual feasible** in the new dual problem. Since both the primal solution x^* and the dual solution $(y^*, 0)^T$ are feasible, and the duality gap is still zero (as $a_{m+1}^T x^* - b_{m+1} = 0$), the solution x^* **remains primal optimal**.
- Otherwise ($a_{m+1}^T x^* \neq b_{m+1}$): The constraint is **violated**, so x^* is **no longer primal feasible**. However, since the dual feasibility constraints only depend on the constraint matrix A and the cost vector c , the solution $(y^*, 0)^T$ **remains dual feasible**. Since we have a dual feasible starting point that is primal infeasible, we can initiate the **Dual Simplex method** to restore feasibility and find the new optimum.

Motivation & Benefits of Sensitivity Analysis

The main goal of sensitivity analysis (or post-optimality analysis) is to **evaluate small changes** to the LP model without having to solve the entire problem again from scratch. The primary benefit is computational efficiency and rapid decision-making.

Analysis for Adding a New Variable (x_{n+1})

Motivation: To quickly assess a **new opportunity** (e.g., launching a new product or adopting a new technology) by checking if it improves the existing optimal solution.

Benefit: This process is highly efficient because the check requires only the calculation of the reduced cost ($\bar{c}_{n+1}$) of the new variable, using the previously obtained optimal dual solution (y^*):

- If $\bar{c}_{n+1} \geq 0$, the current solution remains optimal, and the process stops immediately.
- If $\bar{c}_{n+1} < 0$, **the new opportunity is profitable**. We know the current solution is primal feasible but **dual infeasible**, so we initiate the Primal Simplex method, **which is often faster than starting the solution process anew**.

2. Analysis for Adding a New Constraint

Motivation: To quickly determine the impact of a new constraint (e.g., a new regulation or reduced resource availability) on the optimal plan.

Benefit: This process quickly selects the correct Simplex path based on primal feasibility:

- We first check if the existing optimal solution (x^*) **satisfies the new constraint**. If it does, x^* remains optimal.
- If x^* violates the new constraint, the solution is **primal infeasible** but remains **dual feasible** (as the dual variables y satisfy the original dual constraints). We can then immediately initiate the Dual Simplex method, which is **designed to restore primal feasibility** by working from a dual feasible starting point.

Global/Local Sensitivity Analysis

Consider the problem:

$$\begin{aligned} \min_x \quad & -5x_1 - x_2 + 12x_3 \\ \text{s.t.} \quad & 3x_1 + 2x_2 + x_3 = 10 \\ & 5x_1 + 3x_2 + x_4 = 16 \\ & x \geq 0 \end{aligned}$$

The optimal solution is $\hat{x} = (2, 2, 0, 0)^T$

- Suppose we change a_{11} from 3 to $3 + \delta$.
- Keep x_1, x_2 as basic variables, let $B(\delta)$ be the corresponding basis matrix.

Your tasks are:

1. Compute $B(\delta)^{-1}b$. For which values of δ is $B(\delta)$ a feasible basis?

Solution:

$$B(\delta) = \begin{bmatrix} 3+\delta & 2 \\ 5 & 3 \end{bmatrix}, \quad B(\delta)^{-1} = \frac{1}{3\delta-1} \begin{bmatrix} 3 & -2 \\ -5 & 3+\delta \end{bmatrix},$$

$$B(\delta)^{-1}b = \frac{1}{3\delta-1} \begin{bmatrix} 3 & -2 \\ -5 & 3+\delta \end{bmatrix} \begin{pmatrix} 10 \\ 16 \end{pmatrix} = \frac{1}{3\delta-1} \begin{pmatrix} -2 \\ -2+16\delta \end{pmatrix}$$

We have to require: $3\delta - 1 < 0 \implies \delta < 1/3$ and $-2 + 16\delta \leq 0 \implies \delta \leq 1/8$.
Hence we must have $\delta \leq 1/8$.

2. Compute $c_B^T B(\delta)^{-1}$. For which values of δ is $B(\delta)$ an optimal basis?

Solution:

$$c_B^T B(\delta)^{-1} = \frac{1}{3\delta-1} (-5, -1) \begin{bmatrix} 3 & -2 \\ -5 & 3+\delta \end{bmatrix} = \left(\frac{10}{1-3\delta}, \frac{\delta-7}{1-3\delta} \right)$$

$$y = -(A_B^T)^{-1} c_B = \begin{pmatrix} \frac{-10}{1-3\delta} \\ \frac{7-\delta}{1-3\delta} \end{pmatrix}$$

Dual feasibility is defined by the constraints: $A^T y + c \geq 0$. Let's check if it holds:

$$A^T y + c = \begin{bmatrix} 3+\delta & 5 \\ 2 & 3 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} \frac{-10}{1-3\delta} \\ \frac{7-\delta}{1-3\delta} \end{pmatrix} + \begin{pmatrix} -5 \\ -1 \\ 12 \\ 0 \end{pmatrix} \geq 0 \implies \frac{1}{1-3\delta} \begin{pmatrix} 5-15\delta \\ 1-3\delta \\ -10 \\ 7-\delta \end{pmatrix} \geq \begin{pmatrix} 5 \\ 1 \\ -12 \\ 0 \end{pmatrix}$$

Notice that the first two constraints are tight (near $\delta = 0$). From the third constraint we obtain:

$$\frac{-10}{1-3\delta} \geq -12 \implies \delta \leq \frac{1}{18}$$

1 Another Farkas lemma

This is a variation of the Farkas lemma we show in the lectures. It is concerned with linear inequalities (instead of equalities and nonnegative solutions)

We will prove that exactly one of the following two statements holds:

1. $\exists x$ s.t. $Ax \leq b$
2. $\exists y$ s.t. $A^T y = 0$, $b^T y < 0$, $y \geq 0$

Solution:

- **CLAIM 1:** Both statements cannot be true.

We will prove this claim by contradiction. Assume both statements are true, i.e., $\exists x$ s.t. $Ax \leq b$ **and** $\exists y$ s.t. $A^T y = 0$, $b^T y < 0$, $y \geq 0$. Then

$$\underbrace{(Ax - b)^T}_{\leq 0} \underbrace{y}_{\geq 0} \leq 0 \implies \underbrace{x^T A^T y}_{=0} - b^T y \leq 0 \implies -b^T y \leq 0 \implies b^T y \geq 0$$

which contradicts our assumption that $b^T y < 0$.

- **CLAIM 2:** Both statements cannot be false.

To show this we identify two cases:

- **Case 1:** if $\exists x$ s.t. $Ax \leq b$ then statement 1 is true.
- **Case 2:** if $\nexists x$ s.t. $Ax \leq b$ (that is, statement 1 is false), then we will show that statement 2 is true.

We will use duality theory to show that. Consider the following LP problem and its dual:

$$(P) \begin{cases} \min_x & 0^T x \\ \text{s.t.} & Ax \leq b \end{cases} \qquad (D) \begin{cases} \max_x & -b^T y \\ \text{s.t.} & A^T y + 0 = 0 \\ & y \geq 0 \end{cases}$$

Since we assumed that (P) is infeasible, from strong duality we can derive that (D) can be either infeasible or unbounded (see the table in the slides, with the different possible cases).

We can easily see that (D) cannot be infeasible because $y = 0$ satisfies its constraints.

Then (D) must be unbounded. Since (D) is a maximization problem, its objective function must take arbitrarily large values¹. This implies that there must exist at least one **feasible solution** $\bar{y}$ (i.e., $A^T \bar{y} = 0$ and $\bar{y} \geq 0$) such that $-b^T \bar{y} > 0 \implies b^T \bar{y} < 0$. To summarize, we have shown that the following conditions hold:

$$\bar{y} \geq 0, \quad A^T \bar{y} = 0, \quad b^T \bar{y} < 0$$

Therefore statement 2 is true.

¹The values of the objective function of unbounded maximization problems tend to $+\infty$. Similarly, the values of the objective function of unbounded minimization problems tend to $-\infty$